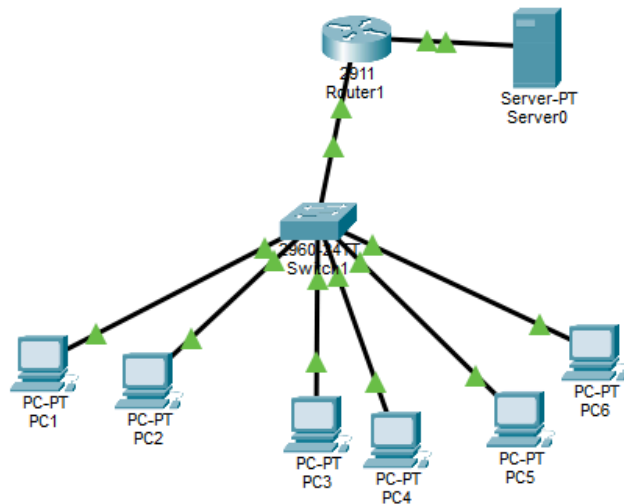


Simulation of Vlan configuration using Cisco packet tracer



CONFIGURATION OF 3 VLAN(ADMIN, SALES & IT SUPPORT) SIMULATION FROM CISCO PACKET TRACER.

QUESTION 1

Security Risk of Not Separating Departments

If departments are not separated using VLANs, all users share the same broadcast domain, which creates several security risks:

- Unauthorized access: Sales staff could access sensitive HR and Finance data belonging to the Admin department.
- Data leakage: Confidential information can be easily intercepted or accessed by unauthorized users.
- Lateral movement: If one PC is compromised by malware, the attacker can easily move to other systems across departments.
- Poor access control: There is no way to enforce departmental security policies.

- Reduced compliance: Financial organizations require network segmentation to meet security and regulatory standards.

How Access Ports Are Configured for Each VLAN

Access ports are used to connect end-user devices (PCs) to the switch. Each access port is assigned to one VLAN only.

Configuration Steps (Example)

Admin VLAN (VLAN 10)

interface range fa0/1 - 2

switchport mode access

switchport access vlan 10

Sales VLAN (VLAN 20)

interface range fa0/3 - 4

switchport mode access

switchport access vlan 20

IT Support VLAN (VLAN 30)

interface range fa0/5 - 6

switchport mode access

switchport access vlan 30

Why Trunk Ports Are Important Here:

- Enable inter-VLAN routing
- Reduce the need for multiple physical connections
- Allow the router to identify VLAN traffic using 802.1Q tagging
- Support secure and scalable network design

Question 2:

Task 1

1 How Inter-VLAN Routing Works

Inter-VLAN routing allows devices in different VLANs (separate broadcast domains) to communicate with each other using a Layer 3 device.

- Each VLAN is assigned a unique IP subnet
- A router or Layer 3 switch provides a gateway IP address for each VLAN
- When a device wants to communicate with another VLAN, traffic is sent to the default gateway
- The Layer 3 device routes the traffic to the destination VLAN based on its routing table

In this scenario:

The router receives tagged VLAN traffic from the switch, routes it between VLANs, and enforces security policies.

2 Where Inter-VLAN Routing Is Configured

Inter-VLAN routing can be configured on:

- A Router (Router-on-a-Stick): Router, because it is cost-effective and suitable for small to mid-sized branch offices.

3 One Rule to Allow Admin but Restrict Sales

Admin handles sensitive HR and Finance data, while Sales has no business need to access payroll systems.

Task 2: Protecting Sensitive Financial Systems

1 What is an ACL? and Why It Is Important:

An Access Control List (ACL) is a set of rules used to permit or deny network traffic based on:

- Source IP address
- Destination IP address
- Type of traffic

Importance of ACLs:

- Enforces least privilege access
- Protects sensitive financial systems
- Prevents unauthorized access between departments
- Reduces attack surface and lateral movement
- Supports compliance and security policies

2 Simple ACL Logic (No Syntax Required)

ACL Policy Logic:

- Allow IT VLAN to access all VLANs and cloud resources
- Deny Sales VLAN access to Admin VLAN resources
- Allow Admin VLAN access to cloud payroll system
- Allow all other legitimate traffic

3 Where the ACL Should Be Applied

The ACL should be applied on the router's VLAN interface, specifically:

- Inbound on the router interface connected to the VLANs

Reason:

- Traffic is filtered before it reaches sensitive systems
- Centralized security control point
- Prevents unauthorized traffic from crossing VLAN boundaries